Yash Kataria

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EDUCATION

Bachelor of Engineering, Hong Kong University of Science and Technology (HKUST) **2023 - 2027**

- BEng in **Computer Science** with a Minor in **Actuarial Mathematics**
- HKUST Beyond Academic Admissions Scholarship | Coverage: 2 years
- Relevant Coursework: Machine Learning, System Optimization, DSA, Software Engineering, DL

TECHNICAL SKILLS

Programming Languages: C++, Python, Java, C, HTML, JS
Developer Tools: AWS, Azure, Xcode, MS Visual Studio, IntelliJ, GitHub
Database Management: SQL, MongoDB, DAX, Power BI
https://ibofferbriefing.youcanbook.me/ Languages: English, Mandarin, Hindi

PROFESSIONAL EXPERIENCE

BNP Paribas **July 2025 – January 2026**

Software Engineer - Automated Market Making Trainee – Global Markets

- Develop Python 3.10, C++ 17 scripts to automate data capture, process and regen. Decreased pipelining latency for Options and Futures Trades by **multi-agent GUI system** via eliminating manpower.
- Developed an Internal **MCP** enabled **RAG**-based Chatbot wrapper aggregating documents, internal databases, key web infrastructures and communications along with semantic retrieval.
- Automate and optimize trading platforms via software engineering, system improvements, and blameless post-mortems.
- Develop/maintain automation tools, alpha models for MMS, backtesting environments, and risk controls for algorithm trading strategies.

QI group; QNet – Hong Kong, HK **June 2024 – August 2024**

Data Analyst and IT Intern

- Sorted company data into insightful visual presentations through classification algorithm using Python data-frames and SQL Queries/DAX on Power BI to spear head financial decisions at board meeting.
- Built a neural network with **97.58% accuracy** as part of a collaborative team to identify business strategies and customer habit analysis saving redundant manpower up to **$50,000 in salary costs**. Refined through back propagation.

The Hong Kong University (HKU) **June 2024 – August 2024**

Research Fellowship Internship

- Developed python algorithm saving **months of work** and **$700 USD monthly** to selectively identify, classify and mass download files from Sec Edgar gov page, enabled easy access to all files of N-PORT data type for Professor's research.
- Used LLM's in Neo4j aura database software to extract data from files manipulated to create knowledge graphs using Cypher and Python used to drive business insights from data.
- Increasing efficiency by up to **2 hours daily** through Restful API companion tools from python to collaborate Gmail and WhatsApp.

Junior Achievement Enterprise - Hong Kong, HK **July 2022 – August 2023**

Director of Technology and Finance

- Co-founded a successful startup, scaled the business to **+400 customers**, increased **revenue by 200% in 2 years**. Directing technology and finance operations.
- Created website, increasing **click-to-conversion rate by 300%.**

PROJECT AND RESEARCH EXPERIENCE

Founder & CEO | Lucida (Assistive Tech) – (2024-present)

Affordable AI-powered Navigation App ecosystem and hardware for the visually impaired

Python, Flask, TensorFlow, OpenCV, React, GPS

- Designed **$1.2M projected** multi-modal smart vest system combining vibration alerts, real-time multilingual audio descriptions (<50ms latency), and computer vision object detection (**98.2% accuracy across 200+ obstacle classes**) hardware solution at **<10%** cost than alternatives ($300 vs market average $5,300). Roughly **40% faster** route navigation vs white cane baseline (HKPU study).
- Secured **$100K HKSTP Ideation Grant** and partnerships with 7 NGOs including Blind Union Asia, enabling beta-testing with 25+ users across 3 countries.

Data Formulator – AI enhanced "Power BI" Platform | Maxims Group Contracted Project

Node.JS, Python, Typescript, JS, HTML

- Developed a Power BI alternative using Microsoft's open-source Data Formulator, integrating AI-generated tables and query-based visualizations, reducing **HKD 900K** in subscription costs.
- Built an AI chatbot onto Data Formulator with end-to-end data connections, enhancing data analysis by **50%**.
- Created a marketing campaign chatbot to answer employee queries on marketing & rewards schemes, leveraging word embedding.

StorageSync – Cross-Platform Inventory Manager

TypeScript, React, Supabase (PostgreSQL + Realtime), Tailwind CSS, Capacitor (iOS & Android)

- Shipped a full-stack household inventory app with recursive location trees, QR/barcode batch-scanning, real-time sync, and a knowledge-graph visualisation; deployed natively to iOS/Android Play Store/App Store and web; Vercel.
- Integrated Strip Subscription for Enterprise and Homeowner use with supabase feature gating.

ContextSign – Context-Aware ASL Fingerspelling Recognition

Python, MediaPipe, scikit-learn, NLTK, OpenCV

- Built a 3-layer real-time ASL fingerspelling system: KNN/SVM classifier over 89-dim hand-landmark features → trigram LM (Brown Corpus) to resolve low-confidence predictions → TTS audio accessibility.

Pathfinder - A* Algorithm

Additional Experience and Awards

- Morgan Stanley – Code to Give | Second Runner up.
- HKSTP Techathon+2025 (Merit) | Conditional for Ideation Programme cum Grant.
- Cathay Pacific Hackathon | Semi-Finalist 2024.